

AeroProf Intelligence: SGA Angola Proof of Concept (PoC)

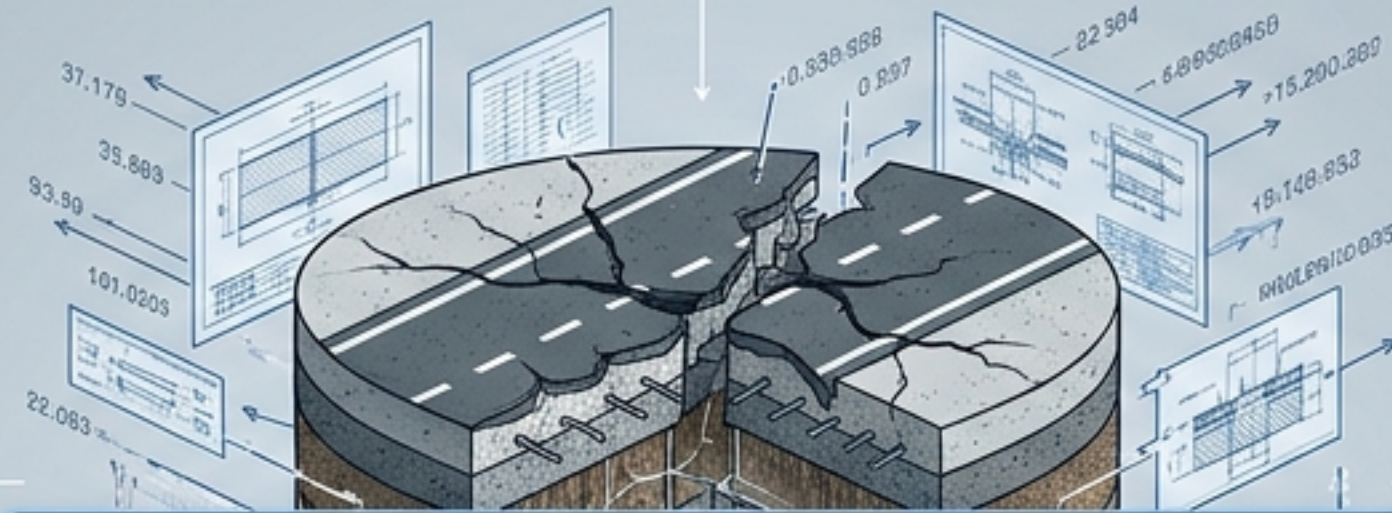
Structuring the eMaintenance paradigm for
airport infrastructure.

Target: Sociedade Gestora de Aeroportos (SGA)
Objective: Validate integrated surface, subsurface, and
structural data fusion.
Framework: ACR/PCR Compliance & Decision Intelligence.

THE FRAGMENTED PRESENT

Decision-making currently relies on disconnected spreadsheets, isolated visual inspections, and outdated as-built drawings.

DATA CORE



THE OPERATIONAL RISK

Without traceability, it is impossible to accurately prioritize network-wide maintenance or confidently clear heavier aircraft for operation.

What is the actual structural capacity of our pavements today, which aircraft can they receive, where are the vulnerabilities, and when must we intervene?



THE MISSING LINK

We lack a unified spatial reference. Surface defects, subsurface anomalies, and structural strength are measured in silos.

The Regulatory Catalyst: ICAO Mandate Nov 2024

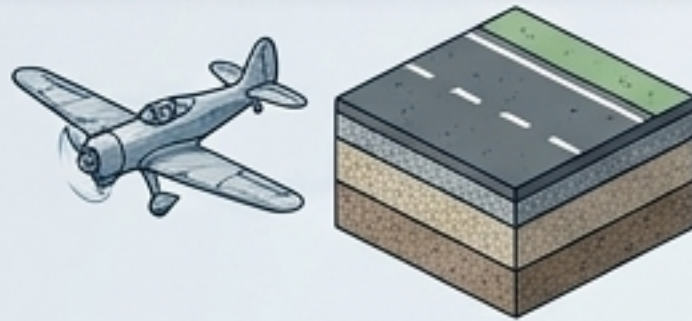
By November 2024, the global standard shifts from ACN/PCN to ACR/PCR.

DEADLINE 

ACN/PCN - The Past

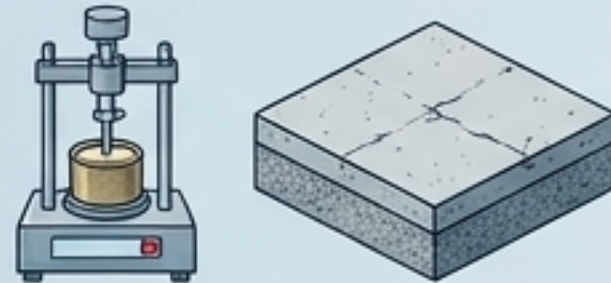
Method:

Empirical (1930s/40s Boussinesq/Westergaard theories).



Parameter:

CBR (Flexible) or k reaction modulus (Rigid).



Flaw:

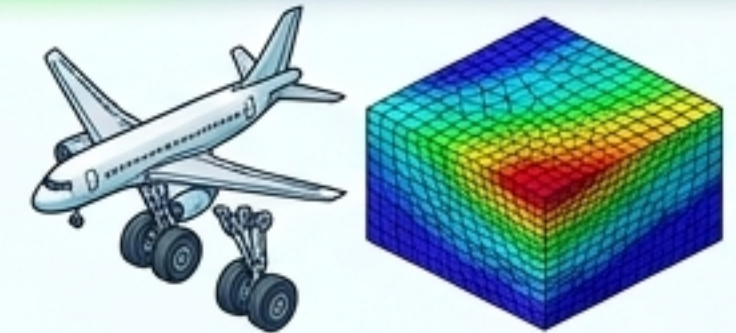
Overly conservative, unable to accurately model complex modern landing gear configurations.



ACR/PCR - The Future

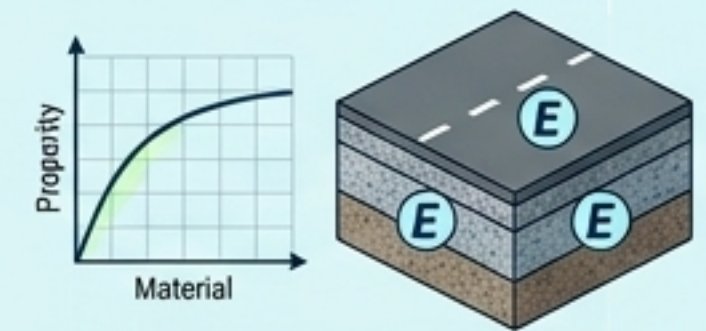
Method:

Mechanistic-Empirical (Elastic-linear or finite element analysis).



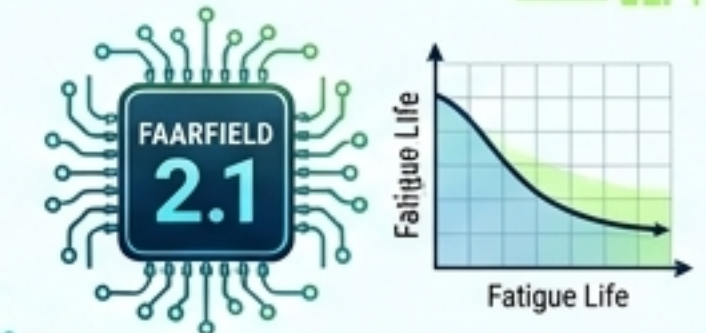
Parameter:

Elasticity Modulus (E) unified for both flexible and rigid pavements.



Engine:

FAARFIELD 2.1 (Optimized asset lifecycle and accurate fatigue modeling).



Demystifying CDF (Cumulative Damage Factor)

Damage is not determined purely by aircraft weight. It is a function of gear configuration, load distribution, and fatigue over time.

0

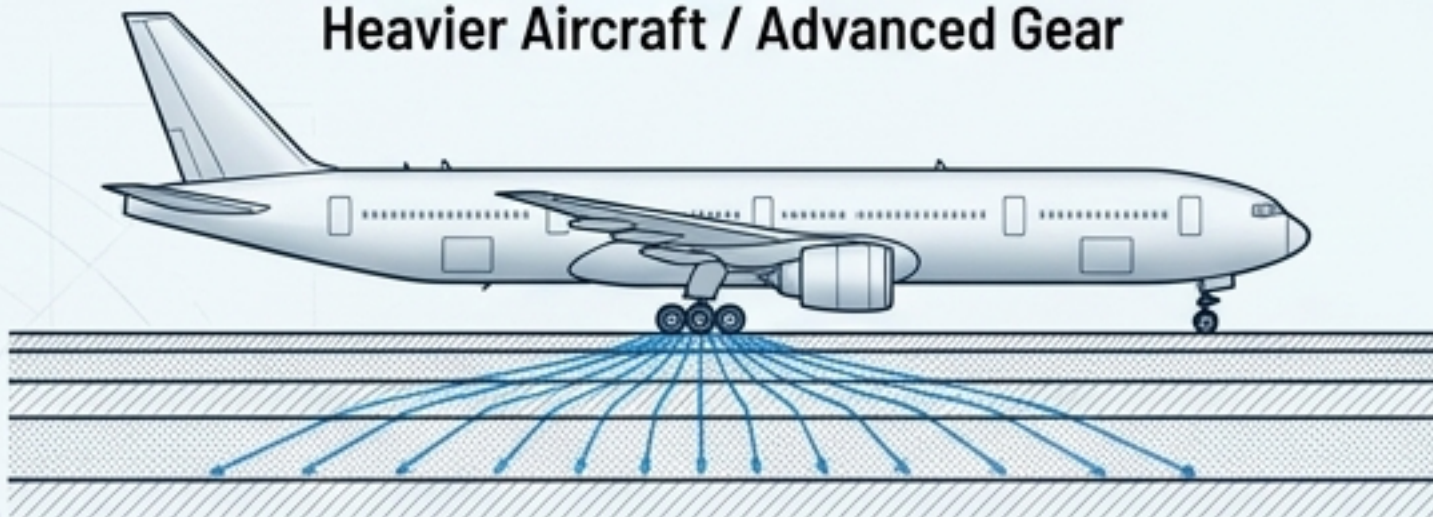
New structure,
zero fatigue

1

Consumed design life
(e.g., rutting, structural failure)

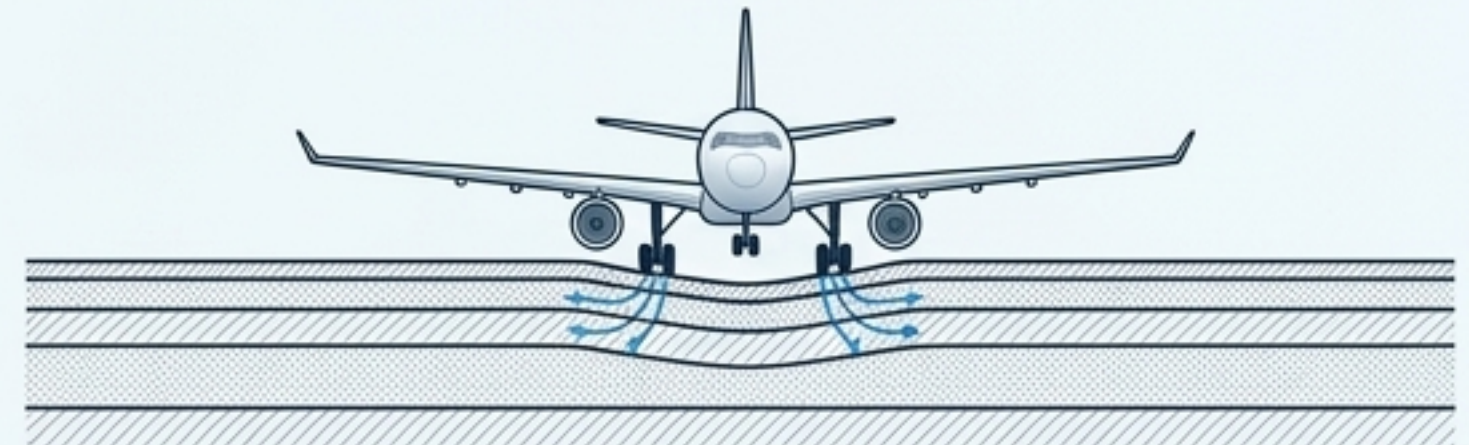


Heavier Aircraft / Advanced Gear



Spreads the load
Spreads the load (lower CDF impact per pass).

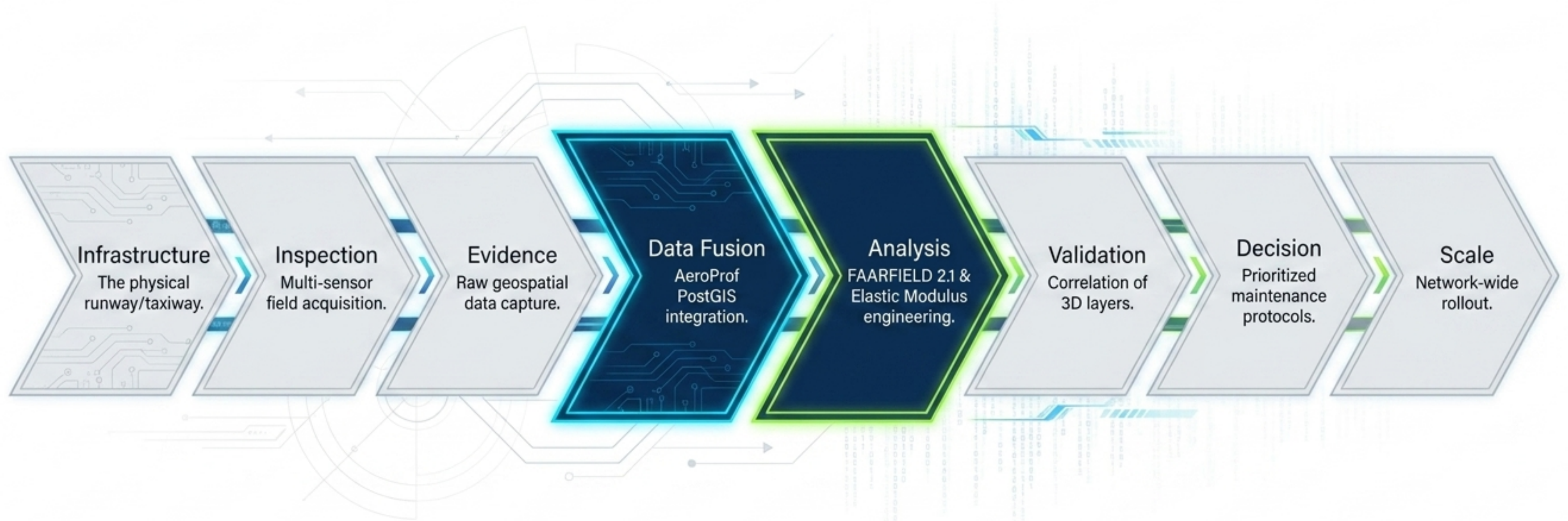
Lighter Aircraft / Concentrated Gear



Concentrated gear causes higher stress
Concentrated gear causes higher stress (higher CDF impact per pass)

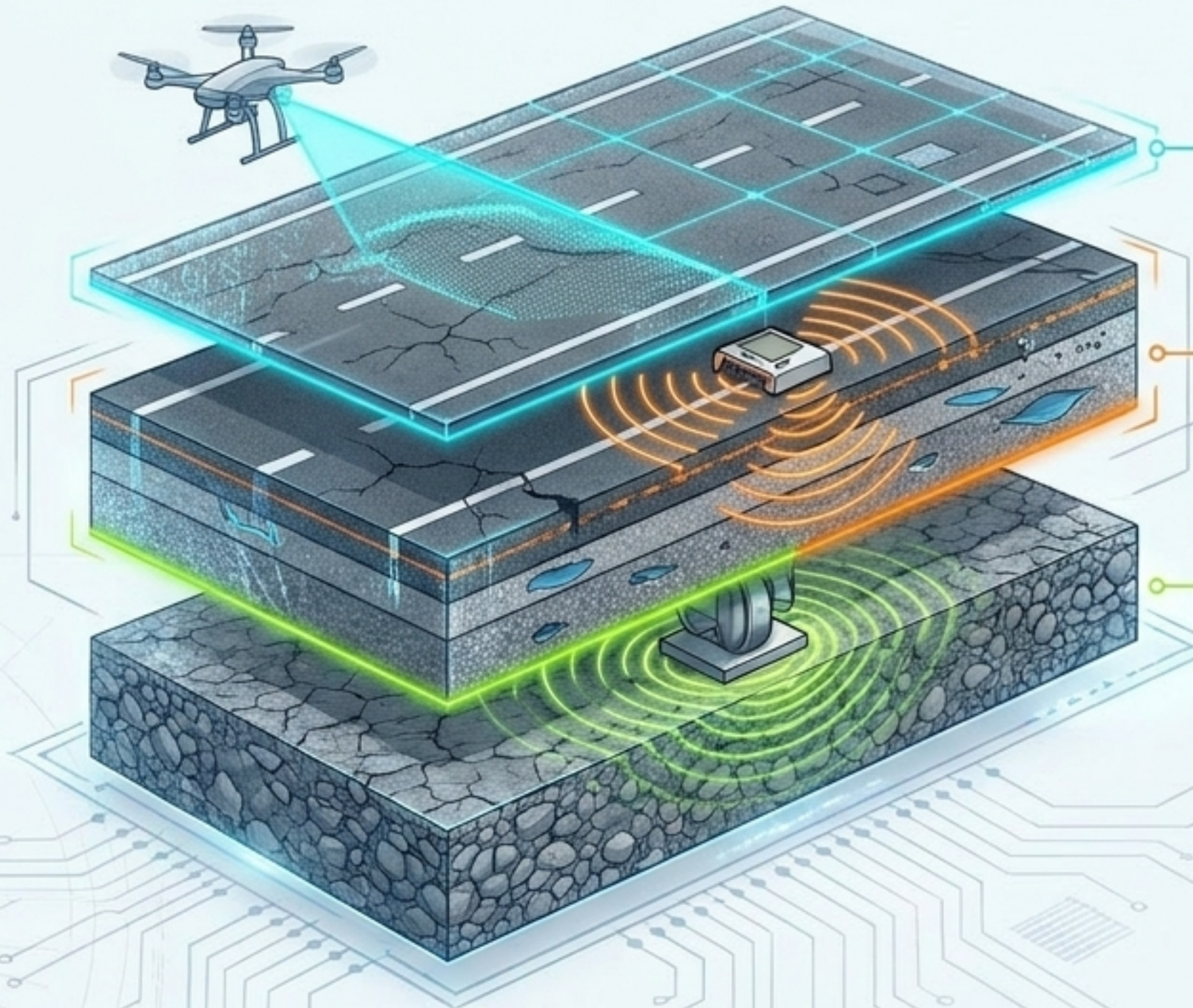
PREMIUM ENGINEERING DOSSIER

The AeroProf eMaintenance Paradigm



The Technological Matrix: Fusing 3 Physical Layers

The Rule: We never declare structural capacity based purely on surface or GPR data. True PCR requires all three layers perfectly aligned in one coordinate system.



Layer 1: Surface

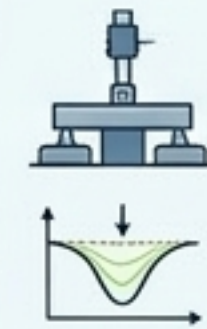
Drone RTK/LiDAR

Capturing geometric truth, visual condition, and 3D point clouds.



Layer 2: Subsurface

GPR (Ground Penetrating Radar). Identifying layer interfaces, thicknesses, and hidden anomalies (water, voids).



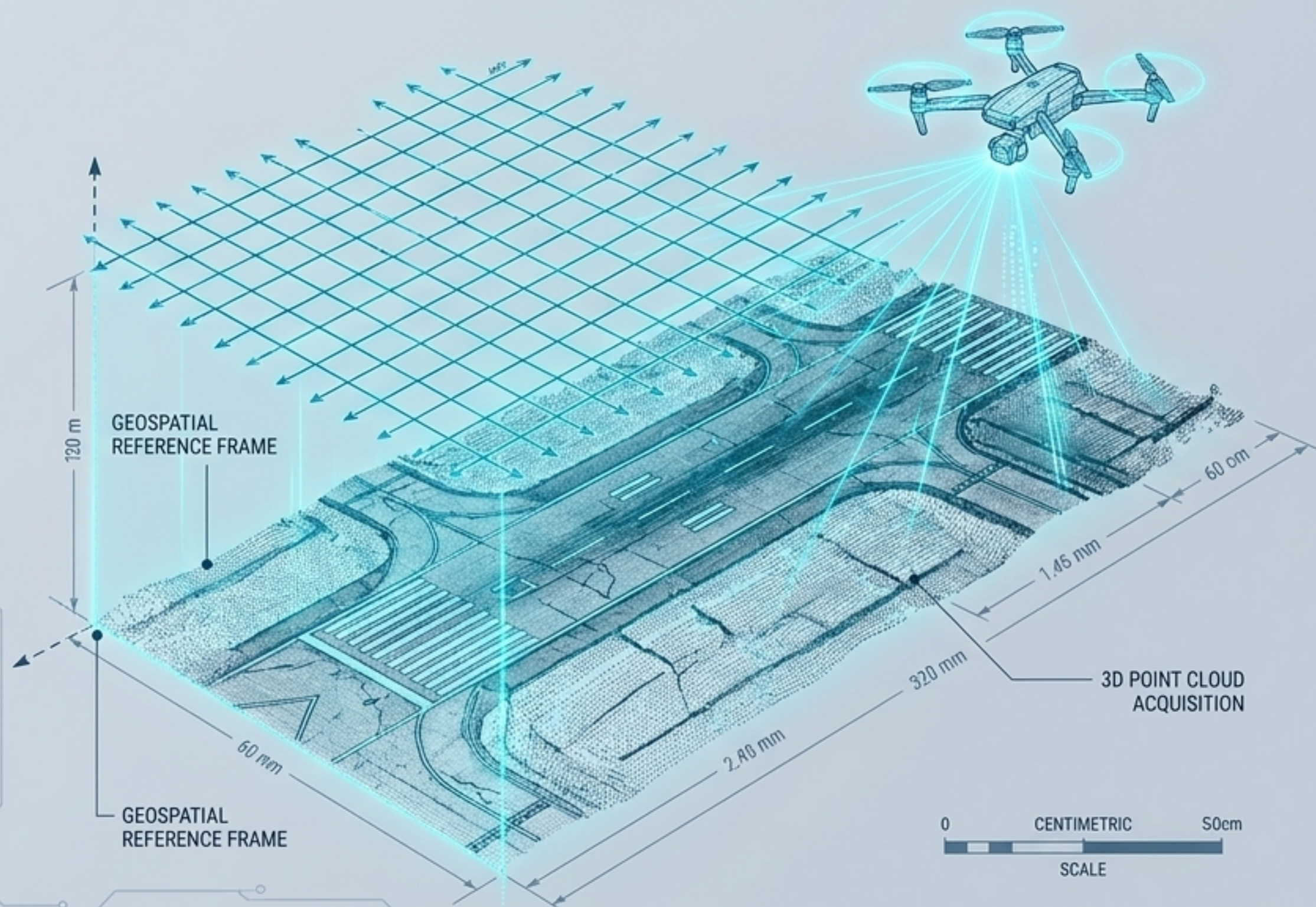
Layer 3: Structure

HWD (Heavy Weight Deflectometer).

Capturing dynamic load response and deflection basins.

Layer 1: Surface Characterization (Drone RTK/LiDAR)

Provides the foundational geospatial mesh. Every subsequent subsurface and structural reading will be pinned to this centimetric coordinate system.



Telemetry Dashboard



Technology

UAS RTK/PPK + LiDAR
+ High-Res RGB (e.g.,
DJI Enterprise).



Precision Target

5cm horizontal and 4cm
vertical accuracy.



Outputs

Orthomosaics, LAS/PLY/PCD
3D point clouds, geometric truth



Layer 2: Subsurface Investigation (Airborne GPR)

Continuous profiling to estimate layer thicknesses and detect discontinuities, moisture, or voids before they collapse into surface failures.

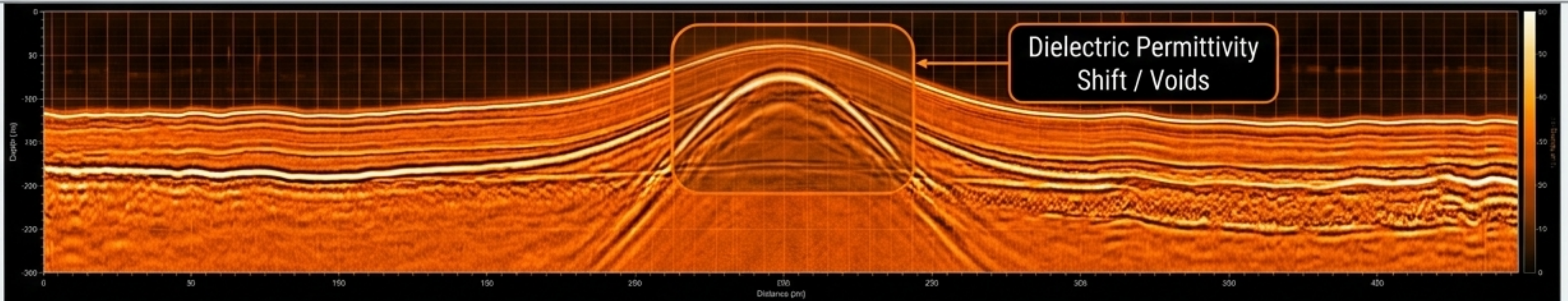
Hardware Context



Hardware Specs

- **System:** Zond Aero LF GPR System.
- **Antennas:** 100 / 150 / 300 MHz (tailored to pavement material and resolution).
- **Penetration:** Up to 12 meters deep.
- **Data:** Standard Geophysical SEG-Y Format.

Data Output Context



Layer 3: Structural Response (HWD)

These precise deflection measurements are the mandatory raw inputs fed into FAARFIELD 2.1 to back-calculate the subgrade's Elasticity Modulus and generate the final PCR.

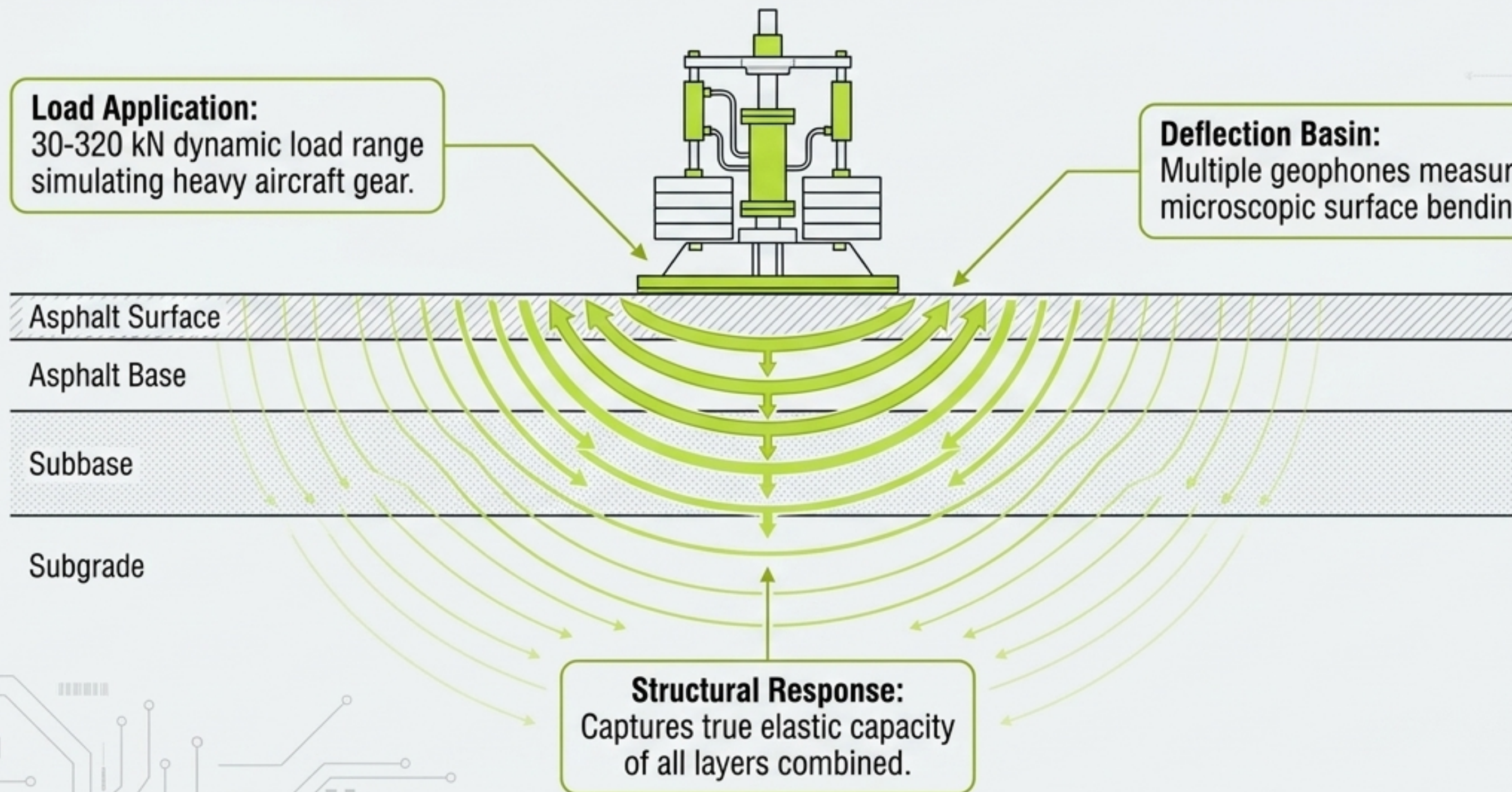
The Anchor: HWD (Heavy Weight Deflectometer) is the absolute core of the structural evaluation. GPR and LiDAR cannot measure load capacity.

Load Application:

30-320 kN dynamic load range
simulating heavy aircraft gear.

Deflection Basin:

Multiple geophones measuring
microscopic surface bending.



The Data Fusion Engine (AeroProf Intelligence)

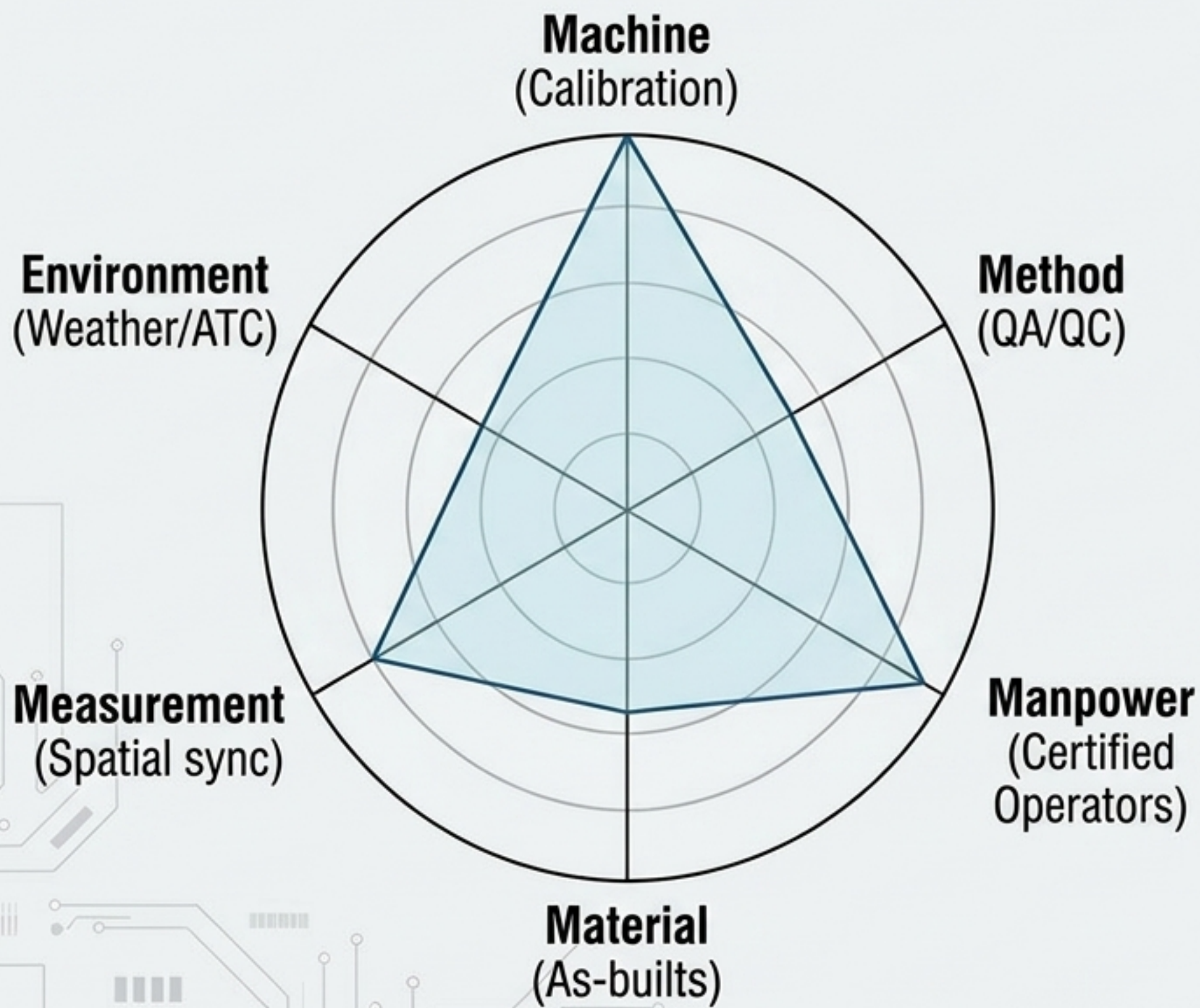


PoC Governance & RACI Framework

Stakeholder		Responsibilities (RACI)
	SGA (Airport Operations)	Accountable for Airside access, providing as-built/historical traffic data, and ATC/NOTAM coordination. (A)
	DWELL (AeroProf)	Responsible for software integration, spatial database architecture, data ingestion, and final dashboard delivery. (R)
	Local Partners / Operators	Responsible for physical hardware operation (HWD, GPR, Certified UAS pilot). (R)

R=Responsible, A=Accountable, C=Consulted, I=Informed

Managing Risk: The 6M Readiness Gates

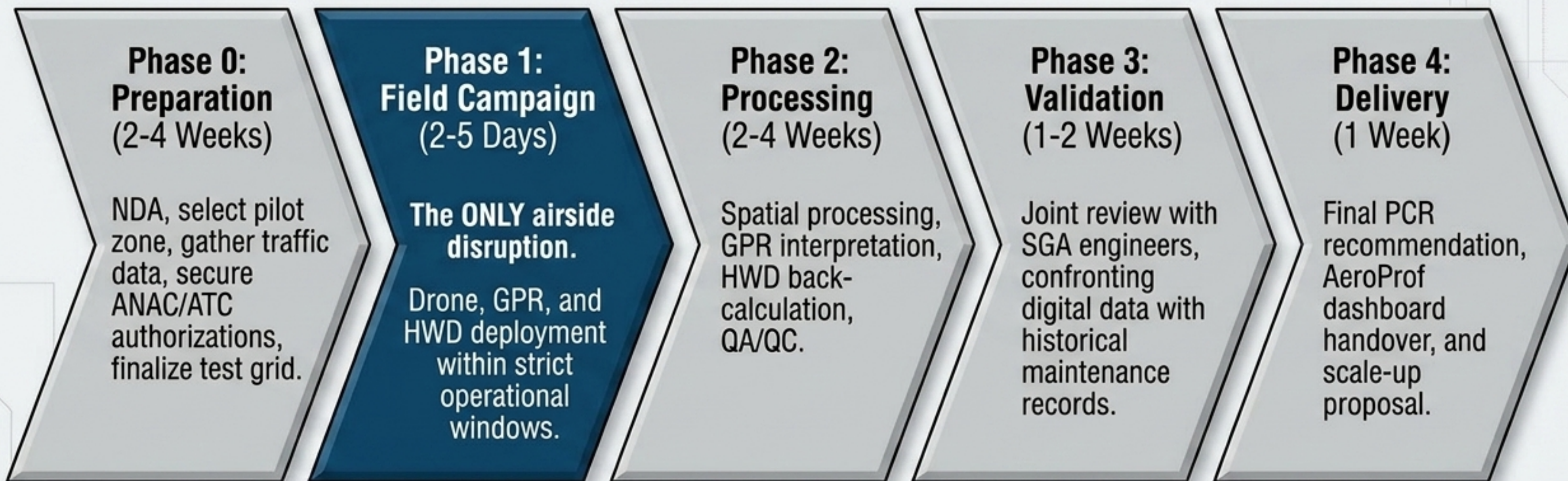


Critical Gate (NTA 36 / NTA 10)

Status: Mandatory Block.

- No drone flies without explicit ANAC Angola explicit ANAC Angola UAS authorization, airspace coordination, and local operator licensing.
- GPR/HWD terrestrial surveys can proceed independently while UAS is cleared.

The Execution Timeline



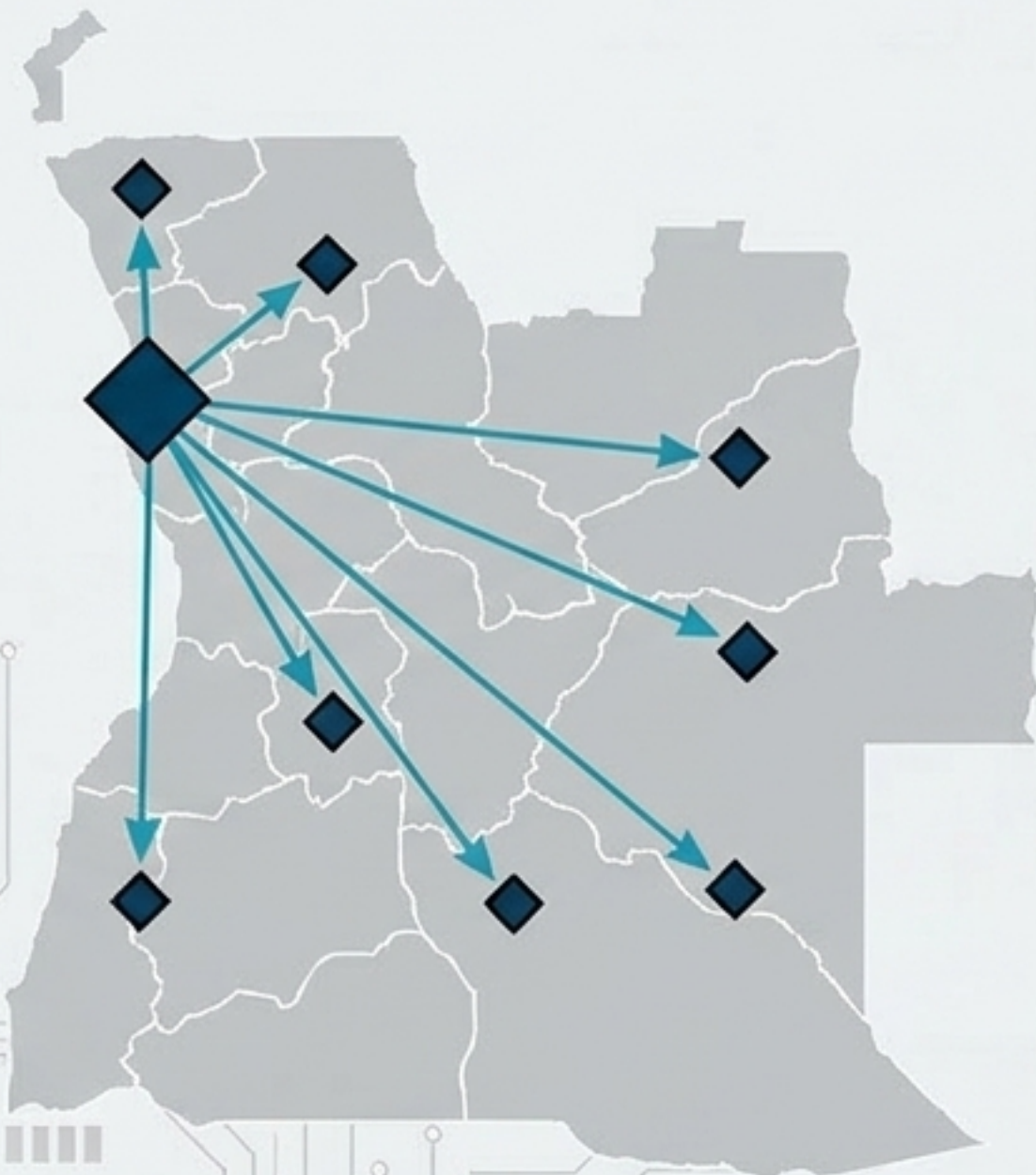
Defining PoC Acceptance Criteria

- ✓ **Criterion 1:** 100% of critical datasets (Surface, GPR, HWD) perfectly aligned in one geospatial reference frame.
- ✓ **Criterion 2:** Unbroken traceability from the raw sensor data file to the final AeroProf dashboard recommendation.
- ✓ **Criterion 3:** Validated execution of FAARFIELD ACR/PCR methodology using integrated traffic and structural inputs.

Strict Engineering Boundary: We will never draw conclusions about structural resistance based purely on GPR or LiDAR. Structural claims require HWD and traffic data.

Beyond the Pilot: Scaling the eMaintenance Network

Once validated, this PoC becomes the blueprint. A standardized, mathematically rigorous method to evaluate, monitor, and optimize every pavement in the SGA network.



Immediate Next Steps for Executive Decision

1. Select the specific Runway/Taxiway pilot zone.
2. Provide baseline historical traffic and as-built data.
3. Approve the PoC Technical Plan and mobilize local partners.